

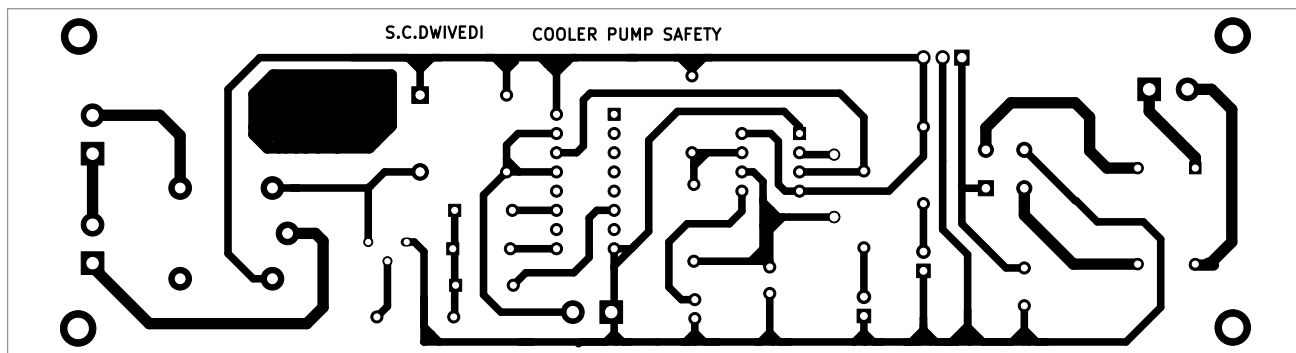


Fig. 3: Actual-size PCB layout

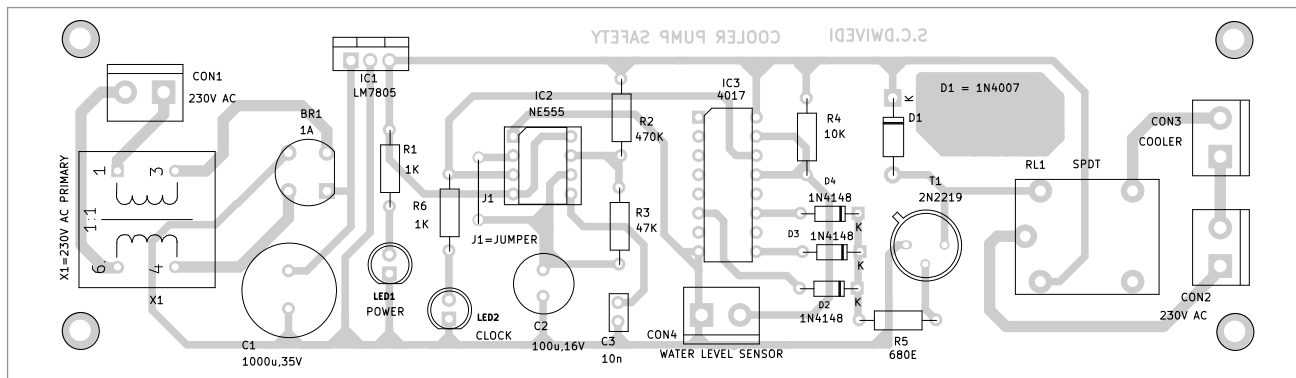


Fig. 4: Component layout of the PCB

water is insufficient, the junction of pins 13 and 15 goes high. In this case, IC2 switches off the cooler pump so that it does not run dry.

PARTS LIST

Semiconductors:

IC1	- LM7805, 5V voltage regulator
IC2	- NE555 timer
IC3	- 4017 decade counter
BR1	- 1A bridge rectifier
LED1, LED2	- 5mm LED
D1	- 1N4007 rectifier diode
D2-D5	- 1N4148 signal diode
T1	- 2N2219 NPN transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R6	- 1-kilo-ohm
R2	- 470-kilo-ohm
R3	- 47-kilo-ohm
R4	- 10-kilo-ohm
R5	- 680-ohm

Capacitors:

C1	- 1000 μ F, 35V electrolytic
C2	- 100 μ F, 16V electrolytic
C3	- 10nF ceramic disk

Miscellaneous:

CON1-CON4	- 2-pin connector
S1	- Water sensor probes made of steel (2)
RL1	- 5V, 1C/O relay
X1	- 230V AC primary to 9V, 500mA secondary transformer
	- Cooler pump

The six outputs Q1, Q2, Q3, Q4, Q5, and Q6 of IC3 are not used. Its remaining three outputs Q7, Q8, and Q9 are connected to the anodes of respective 1N4148 diodes D2 through D4. The cathodes of D2 through D4 diodes are connected to each other and to the base of transistor T1 through resistor R5. Only Q7, Q8, and Q9 output pins of IC 4017 are used to drive the relay to increase the off time of the pump. The pump remains off when the seven outputs Q1 through Q6 are high.

Construction and testing

Complete the connections and power on 230V AC. Relay RL1 is driven by transistor T1 when outputs Q7, Q8, and Q9 of IC2 go high to activate the cooler pump. The pump will run only when water is available in the cooler's water storage tank, which is sensed by the sensor probe to enable IC2. The motor will run for around 1/3rd time and remain off for 2/3rd time.

An actual-size, single-side PCB layout for the cooler pump's safety circuit is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on PCB, enclose it in a suitable box. Fix LED1 and CON4 on front side of the box and mains connector's CON2 and pump connector CON3 on the rear side.

Take water sensor probes made of steel rods, as they should be corrosion-free, and connect them to CON2. The other terminals of sensors should be fixed to the cooler's body in such way that distance between the probes is 30mm to 40mm and they get shorted when there is adequate water. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/live-diy-cooler-pump-safety>

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